

A PRINT AND SCAN OPTIMIZED WATERMARKING SCHEME

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Abstract -

We are presenting an algorithm for watermarking digital images with the aim of obtaining a very high quality of the watermarked image and a strong resistance to print and scan processes. We propose a method which combines an additive watermarking algorithm in the spatial domain providing resistance against cropping and exhaustive search and a synchronisation template in the Fourier domain providing resistance against geometrical deformations. The additive watermark in the spatial domain is based on an original generalized 2-D cyclic patterns for secret message embedding. The cyclic property and pattern redundancy facilitate detection and synchronisation against cropping and image processing basic attacks (like compression, filtering, blurring). This algorithm has to be complemented by a template insertion for getting resistance against rotation and scaling which are caused by print and scan processes. We generate the template in the Fourier domain inserting some points locally. The watermark and the template are weighted by a Human Visual masking function. The global scheme, though very classical in its global concept, provides a very efficient protection to digital image which could be delivered both in a digital high quality format and in a printed form.

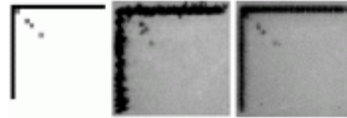
OVERVIEW

Specifications of watermarking schemes depend on their usage scenario. Often, the design of a watermarking algorithm focus on specific attacks : blur, sharpening, compression, noise insertion leads to requirements which are quite different than those that are required for getting resistance against geometric attacks like rotations, stretching, cropping... It is very difficult to hold out against both kinds of attacks (stirmark [1] for example). Often, domain insertion and resistance have an important relationship. Hence, we can divide watermarking algorithm in two categories:

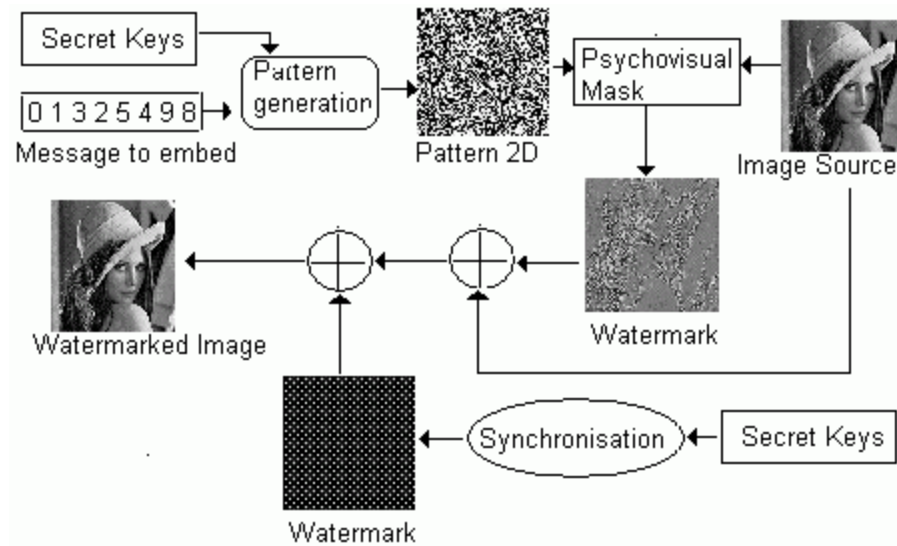
- the message is embedded in a transformed domain (Fourier [2], wavelet [3],...)
- the message is embedded in the spatial domain [4].

We propose in this paper to develop a global method for embedding image data, using both spatial and Fourier domains. We have tried to use the specificity of these two domains. This approach is followed by several research groups (including Digimarc, University of Geneve[8], University of Firenze ...) and suffers from weaknesses due to the additive structure of the watermark (the so-called template and transposition attacks). A strong goal of our algorithm is to resist against print and scan operations.

In fact, print and scan leads to the following list of watermark modifications: scaling, rotation with blur effect and some MIRE default produced by a basic printer.



We propose to develop a private watermarking. It means that the embedding and extraction keys are the same and must be kept secretly as opposed to an asymmetric public watermarking system [5]. In this embedding scheme, three elements are im-



portant: the psychovisual mask, pattern generation, synchronized block. In the following sections, we describe each element of the scheme.

PSYCHOVISUAL MASK

An efficient watermarking algorithm has to combine invisibility and robustness [6]. Robustness is guaranteed by the redundancy of the insertion scheme and invisibility by a psychovisual mask. The purpose of this mask is to modify the watermark according to the image energy to make it invisible. Our psychovisual mask is based on two principles:

- image activity (local mean) which compares pixel intensity inside its environment. In fact, if we increase or decrease some pixel intensity in high contrast region, we can not detect a difference between two pixels.

- importance of pixel intensity. It is easier to see a pixel modification in black intensities than in white intensities. This property is well defined and explained by the Weber-Fechner law[7]

The visual increment threshold is defined as the amount of light ΔB_T necessary to add to a visual field of intensity B to become visible. This threshold can be approximated by piecewise linear functions and the minimum amount can be computed as

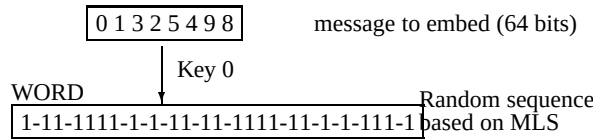
$$\begin{aligned}
 &\text{low intensities region:} \\
 &\Delta B_T = \sqrt{x_1 x_2} * \beta * \sqrt{B} * \left(\frac{\Delta B}{B}\right)_{max} \quad \text{for } B \leq x_1 \\
 &\text{De Vries-Rose region:} \\
 &\Delta B_T = K_2 * \sqrt{B} \quad \text{for } x_1 \leq B \leq x_2 \\
 &\text{Weber region:} \\
 &\Delta B_T = K_1 * B \quad \text{for } x_2 \leq B \leq x_3 \\
 &\text{Saturation region:} \\
 &\Delta B_T = K_3 * B^2 \quad \text{for } B \geq x_3
 \end{aligned}$$

x_1, x_2, x_3 determine the boundaries of the different regions and K_1, K_2, K_3 are constants of proportionality.

These two principles of perceptual model compute a threshold. Under this threshold the watermark is supposed to be invisible.

2-D CYCLIC PATTERNS

To reduce complexity and increase detectability of the watermark, we use Maximum Length Shift sequences [9]. MLS sequences are pseudo-random sequences well known in signal processing. They have simple cyclic properties. Most problems found in watermarking are related to synchronisation. This issue can be resolved by using a cyclic code. Any attacks against MLS sequences is equivalent to a shift in the original sequence. By cross-correlating estimated and original sequences the watermark can be easily detected. To create a efficient and secure algorithm we need to "hide" this correlation. We use therefore two random keys for that purpose. A pseudo random sequence shifted by a key K0 can define a determinist random sequence. This new method based on MLS sequence is developped by D. Delannay [10].

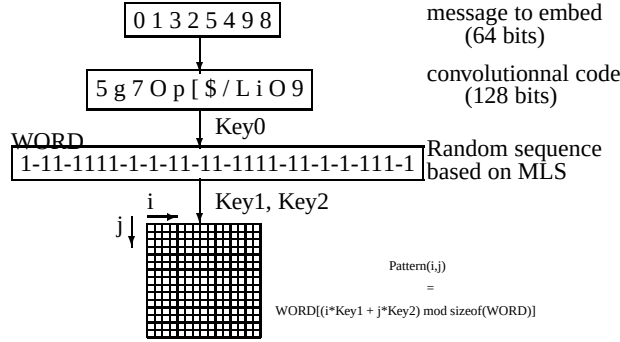


We want to create a 2-D cyclic pattern. So we are going to expand our 1-D cyclic sequence into a matrix. The method used to map 1-D cyclic sequence onto a matrix according to message to embed is called M-ary Modulation [11]. The created tiles are superimposed to the original image with respect to two keys K1 and K2. Hence, the same information embedded in the tiles describes a translation in the tiles.

$$\text{Pattern}(i,j) = \text{WORD}[(i * \text{Key1} + j * \text{Key2}) \bmod \text{length}(\text{WORD})]$$

An error correcting code [12] is added on the 1-D information. We have chosen convolutional code to encode our information and soft Vitterbi [13] to decode the corrupted 1-D signal. The message to modulate is two times the original message.

The final scheme we propose is presented below:



SYNCHRONIZED BLOCK

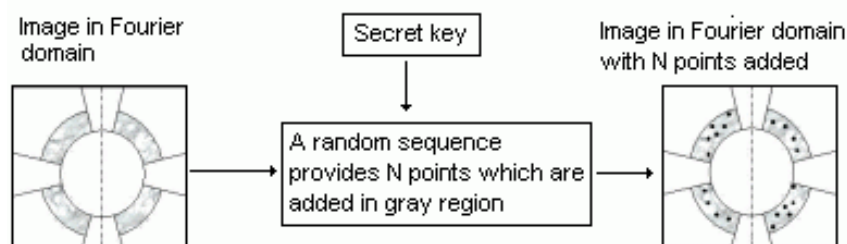
As we have explained before, we need to create a re-synchronisation process since the watermarked image will suffer from rotations and scalings due to the print and scan process. In the spatial domain, it is difficult to elaborate such a process. Some useful mathematical properties can be obtained in the Fourier domain. A scaling transformation in the spatial domain corresponds to a scaling with an inverse factor in the Fourier domain.

$$\begin{aligned}
 TF(f \circ S(S_x, S_y))(u, v) &= \alpha \int_{R^2} f(S_x \cdot x, S_y \cdot y) e^{-(ux+vy)} dx dy \\
 &= \alpha \int_{R^2} f(X, Y) e^{-\left(\frac{uX}{S_x} + \frac{vY}{S_y}\right)} dX dY \\
 &= TF(f) \cdot S\left(\frac{1}{S_x}, \frac{1}{S_y}\right)(u, v)
 \end{aligned}$$

A rotation in the spatial domain has the same effect in the Fourier domain.

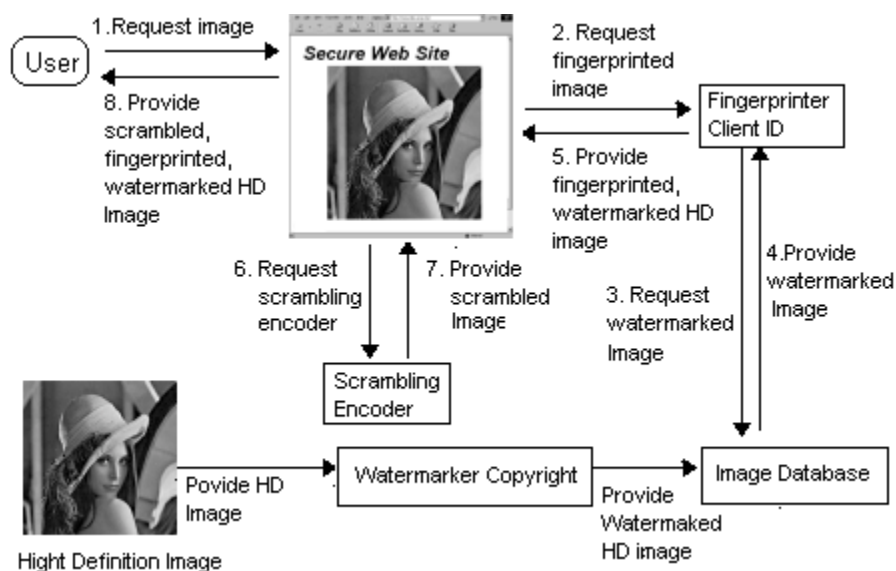
$$\begin{aligned}
 TF(f)(u, v) &= \alpha \int_{R^2} f(x, y) e^{-(ux+vy)} dx dy \\
 &= \alpha \int_{R^2} f(R_\theta(x, y)) e^{-(ux+vy)} dx dy \\
 &= \alpha' \int_{R^2} f(X, Y) e^{-((u, v) \cdot R_{(-\theta)}(X, Y))} dX dY \\
 TF(f \circ R_\theta)(u, v) &= \alpha' \int_{R^2} f(X, Y) \cdot e^{-R_\theta(u, v) \cdot (X, Y)} dX dY \\
 &= \alpha'' TF(f) \cdot R_\theta(u, v)
 \end{aligned}$$

For getting the best compromise between invisibility and robustness we decided to insert the template points in middle frequencies. A key K3 defines with precision this specific region.



APPLICATIONS

The algorithm has been implemented in an IST (European funded project) called ASPIS (An Authentication and Protection Innovative Software System for Dvdrom and Internet). In this project we developed, with other partners, a secure platform around watermarking, fingerprinting and monitoring technologies. During Paris book-fair, we have tested our algorithm with success. Our trial is very simple. A public web site contains images in low resolutions called stamps. When a stamp is selected, users are connected to a secure web site where the image is watermarked with a copyright, fingerprinted with private information and scrambled. Hence, the high resolution image displayed in web browser user or downloaded is well protected.



To test robustness of our watermarking, we have printed the image with a classic inkjet printer and digitized it with a basic scanner. The proposed algorithm has revealed a total reliability in the retrieval of the watermark in the printed documents while the visual quality of the image being preserved

Future work should include some process to resist against direct malevolent attacks like template attacks. Some hiding processes for additive watermarks are under study at UCL.

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